

Developing a Personalized Clinical Support System Utilizing Classification Models for Drug Assessment

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Abstract

This paper presents a clinical decision support system that tailors therapy based on sentiment analysis of patient drug reviews. The study incorporates deep learning with TF-IDF in the NLP modelling to determine whether patients' attitudes toward medicines are positive, negative, or neutral. Then a recommendation system is delivered highlighting the remarkable drugs based on the medical condition of the individuals. Additionally, the paper compares the performance of ensemble methods, deep learning algorithms like LSTM with TF-IDF to traditional machine learning methods like Linear - SVC and logistic regression. The results demonstrate the superior accuracy of 0.97 by the deep learning approach in capturing intricate sentiment patterns, highlighting its potential for enhancing clinical decision systems in healthcare.

Keyword - Natural Language Processing, TF-IDF, Multi-Classification, Sentiment analysis, Supervised Machine Learning, FMCG, Logistic Regression, Drug Reviews, Linear-SVC

1. INTRODUCTION

The development of progressively customized and personalized therapeutic solutions is an ongoing effort in the changing world of modern healthcare. Patients today have the ability to influence healthcare decisions by using a digital platform to communicate their experiences and views [13]. The explosion of internet drug evaluations has resulted in a massive store of patient-generated data, providing a once-in-a-lifetime chance to tap into their thoughts and use this knowledge in clinical decision support systems.

This study takes a risk by attempting to bridge the perception gap between patient experiences and clinician decision-making. This study uses advanced natural language processing (NLP) techniques in conjunction with term frequency-inverse document frequency (TF-IDF) representations to determine patients' opinions regarding medications, categorizing them as positive, negative, or neutral. The goal is to understand the subtle nuances of sentiment patterns in these reviews, laying a solid foundation for the improvement of clinical decision support in the healthcare sector.

The development of a suggestion system that can identify noteworthy medications matched to an individual's specific medical condition is one of the research's main contributions. This technology provides vital help to healthcare practitioners by combining sentiment analysis with medical context, delivering insights into prospective treatment alternatives that correspond with the patient's preferences and requirements.[15]

This paper provides a complete comparative analysis of several machine learning approaches to assure the reliability and efficacy of the proposed strategy. It compares the performance of ensemble methods and deep learning algorithms, particularly Long Short-Term Memory (LSTM) networks combined with TF-IDF, against the performance of standard machine learning techniques such as Linear Support Vector Classification (Linear-SVC) and logistic regression. The findings of this comparison research illustrate the benefits and limits of these various techniques, allowing for more informed decisions about the most successful way for clinical decision assistance.

Preliminary data from this study show that the deep learning approach achieved a stunning accuracy rate of 0.97, demonstrating its ability to comprehend nuanced sentiment patterns buried within drug reviews. This discovery highlights the potential of deep learning approaches to greatly improve clinical decision support systems in the healthcare domain, paving the way for more personalized and patient-centered therapeutic suggestions.

This research contributes to the growing body of literature in the field of sentiment analysis while also presenting a possible route toward more precise and patient-centered healthcare solutions in this journey to use patient sentiments for individualized therapy.

2. LITERATURE REVIEW

Drug review classification is an important aspect in the field of medical clinical trials research and medical industry. It aids in reducing the number of clinical errors based on the previous data typically using the knowledge of Machine Learning and Artificial intelligence algorithms.

The drug review classification involves the use of sentiment classification based on user reviews. Hence addressing the use of text preprocessing with the use of appropriate Logistic regression and SVC algorithms plays a vital role in this research paper to classify sentiments based on user reviews in the paper [1]. However, when it means most of the sentiment classification is based on negative and positive leaving out neutral to be a third classification label which intentionally leaves out the hold of multi label classification which is a drawback in Deep learning models to classify sentiments from social media.

The paper [2], addresses 3 label classification using Recurrent Neural Networks. Sentiments usually revolve around a certain subject or entity from a given review, the usage of aspect-based sentiment classification also makes the classification more meaningful.

The paper [4] addresses the concept of aspect based deep learning classification to categorize the labels based on certain subjects/aspects. However, the accuracy or evaluation metrics do not seem to be promising for certain cases. So, Deep learning models like LSTM or RNN makes the algorithm more accurate for sequential like text/images/videos. Also, semantic similarities are inherited by the usage of LSTM models as mentioned in this paper, which increases the overall performance of the model.

In paper [5] when it comes to the implementation of such sentiment classification, usage of lexicons becomes an important aspect, with the usage of open-source libraries like VADER and TEXTBLOB that can help in more precise labelling and improvement of performance in word embedding techniques like TF-IDF which is used in this research of [6].

As mentioned in this paper [7] the usage of Deep learning technique is a drawback which has been experimented in this research to gain main more insights on the usage of Sequential based deep learning methodology like LSTM (RNN).

3.APPROACH

Dataset:

The University of California, Irvine (UCI) Irene archive data set has been made explicitly available for the investigation of empirical computation utilizing machine learning techniques and existing computation tools. The Drug Review dataset primarily incorporates the patient reviews on specific prescription drugs, as well as information on linked illnesses and a 10 stars patient rating, indicating overall patient satisfaction. There are 215 thousand instances and six attributes in total. It provides summaries of the drug experience and feedback. Out of these 25% is allocated to the testing set and 75% to the training set for learning. The overall depiction of attributes involved is displayed in table 1 below.

Table:1 Dataset Description

Attribute	Datatype	Description
Drug-id	Int	Unique id of each drug
Drug Name	Categorical	Name of the drug
Condition	Categorical	Name of the condition
Review	text	Review by patient
Rating	numerical	10 stars rating highlighting overall satisfaction
Date	yyyy-mm-dd	Review entry date
Useful Count	numerical	Count of users who found useful about drug

Count of condition by usefulCount

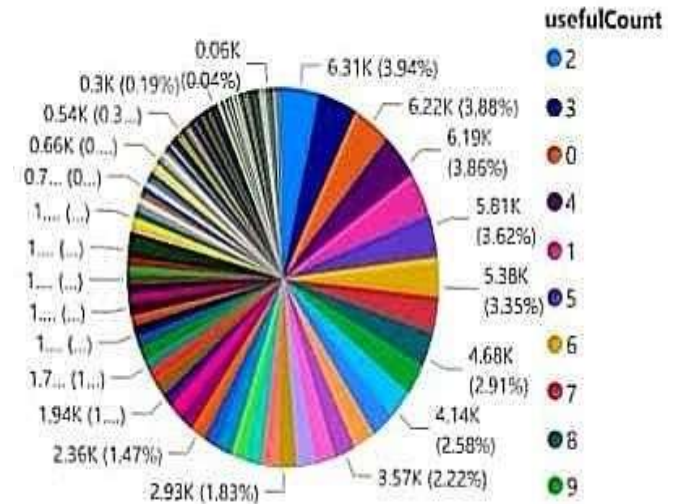
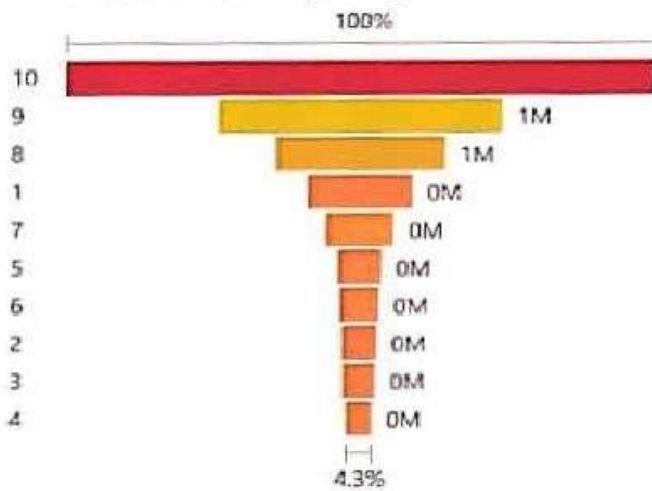


Figure 1: Count of condition by useful Count values

Sum of usefulCount by rating



Count of drugName by rating

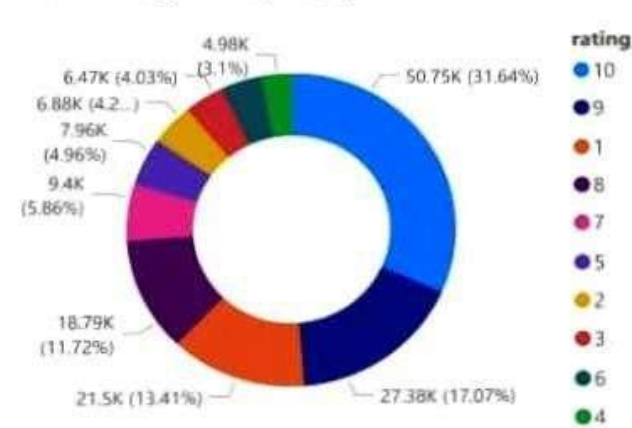


Figure 2: Sum of Useful count and Count of Drug name by Rating

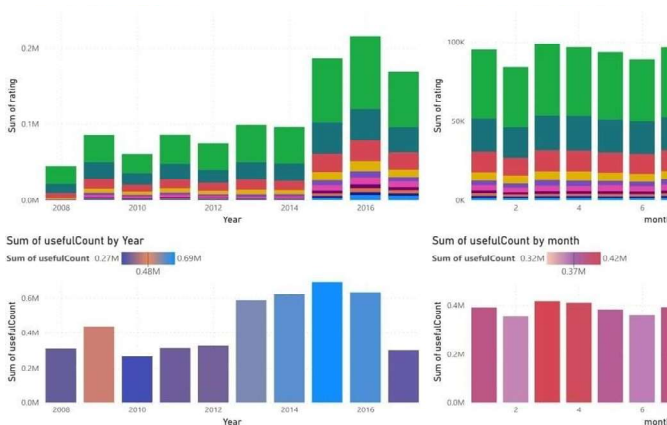


Figure 3: Sum of rating by Year and rating

Sentiment Labeler:

Text blob:

Many natural language processing tasks comprises textual data use the Python library Text-Blob. It provides APIs for many different NLP tasks, such as part-of-speech tagging, sentiment analysis, and noun phrase extraction. It has a vocabulary of 2,918 words for sentiment analysis. The sentiment analysis tool assesses text as either opinion or informational content by assigning polarity and subjectivity scores. It uses the following format to create a tuple for a word or phrase: [6].

$$\text{Sentiment (polarity-score, subjectivity-score)} \quad (1)$$

Vader:

A key lexicon for emotional analysis in the English language is the Valence Aware Dictionary for Sentiment Reasoning (VADER). In order to improve sentiment analysis, it combines qualitative techniques with empirical validation from human raters using a valence-based methodology on a human-centric approach [7]. By considering the valence scores of words contained in its dictionary, it assesses sentiment. Within the polarity range R_p of $[-4.0, +4.0]$, where $+4.0$ denotes a positive sentiment and -4.0 denotes a negative sentiment, these valence scores are represented as floating-point values. It contains a vast lexicon of 7,517 sentiment-related words, including acronyms, slang, and words with valence scores [8]. The output tuple is organized as follows when it is used for sentence analysis.

$$\text{sentiment (negative, neutral, positive, compound)} \quad (2)$$

Word Embeddings:

The weight of a word is based on its relevance to the particular document, and each document uses TF-IDF to quantify the significance of individual words [9]. Inverse Document Frequency (IDF) and Term Frequency (TF) are two metrics combined in this relevance-based computation. While TF evaluates a word's significance within a specific document, IDF measures a word's frequency throughout the entire corpus of texts. In document 'doc,' the word 'wd' has the following mathematical representation of IDF:

$$\text{idf}(\text{wd}, \text{doc}) = \log(\text{No} / (\text{df}(\text{wd}) + 1)) \quad (3)$$

In the equation, the number of documents in the corpus is indicated by the symbol "No," and the number of documents that contain the word "wd" is indicated by the symbol "df(wd)."

An example of the process is given the table below:

Original Review	It has no side effect, I take it in combination of Bystolic 5 Mg and Fish Oil
Lower case conversion and junk removal	it has no side effect, i take it in combination of bystolic 5 mg and fish oil
Stop words removal	no side effect take combin bystol mg fish oil
stemming	no side effect take combin bystol mg fish oil
tokenization	['no', 'side', 'effect', 'take', 'combin', 'bystol', 'mg', 'fish', 'oil']

To compute the TF-IDF of a word wd in document doc following formula can be used:

$$tf\ idf(wd, doc) = tf(wd, doc) \times idf(wd, doc) \quad (4)$$

$$tf\ idf(wd, doc) = tf(wd, doc) \times \log(No / df(wd) + 1) \quad (5)$$

Classifiers:

Machine Learning:

Logistic Regression and the Linear Support Vector Classifier (Linear- SVC) are used to classify the sentiment labels in the reviews. The binary result's log odds are modeled by logistic regression as a linear combination of predictor variables [10]. On the other hand, linear-SVC is a type of supervised machine learning that is used for regression analysis and classification and is acknowledged for performing better than other methods [11].

Deep Learning:

In order to address problems with gradient vanishing or exploding, where gradients can exponentially decrease or increase over long sequences, the Extended Short-term Memory Network (LSTM) was developed [12]. LSTM uses recurrent connections, which set it apart from traditional feedforward neural networks and allow it to analyze entire data sequences, including text, speech, and video, as well as individual data points like images [14]. Together with a cell memory state, the LSTM architecture consists of three essential gates: an input gate, a forget gate, and an output gate. New information must be assimilated into the cell state by the input gate, and information that has become irrelevant or less significant must be removed from the cell state by the forget gate. A crucial element in maximizing the effectiveness of the LSTM network is the multiplication-based filtration process. Finally, the output gate's job is to extract useful data from the current cell state and display it as the output.

3. ARCHITECTURE:

Text Preprocessing/Data Preprocessing:

The acquired data is cleaned in this process, making it the sole data required for sentiment classification. There are various steps in preprocessing such as normalizing the date values, removal of unwanted characters and junk characters is applied further converting the complete text lower case letters. Then, the stop words are removed from the review text which is useful for removing special characters and conjunction words. Finally, the stemming is applied on the dataset to transform all different tense or forms of words into their basic word and tokenizing the stemmed words of the text reviews.

Preprocessing of the text corpus uses the help of the Natural Language Toolkit (NLTK) library with the support of the Powerbi framework. Eliminating stop words involves limiting the dimensions of the term space; the most prevalent words in the text are articles, prepositions, pronouns, and others that do not contribute sense to the document. Then, the words are further split into single terms by the tokenization method to process each word into numeric format. Further, sentiment labelling is performed by using VADER and TEXTBLOB.

Feature selection:

Certain columns are specifically chosen for training and evaluation of the model. This includes the drug-id, drug name, condition, rating, and sentiment labelling of Vader and Text blob. These columns have higher significance as they possess the key idea for the development of the recommendation system. The sentiment labels are used as labels for training the learning models. Others are quite useful in identifying the extent value of each review.

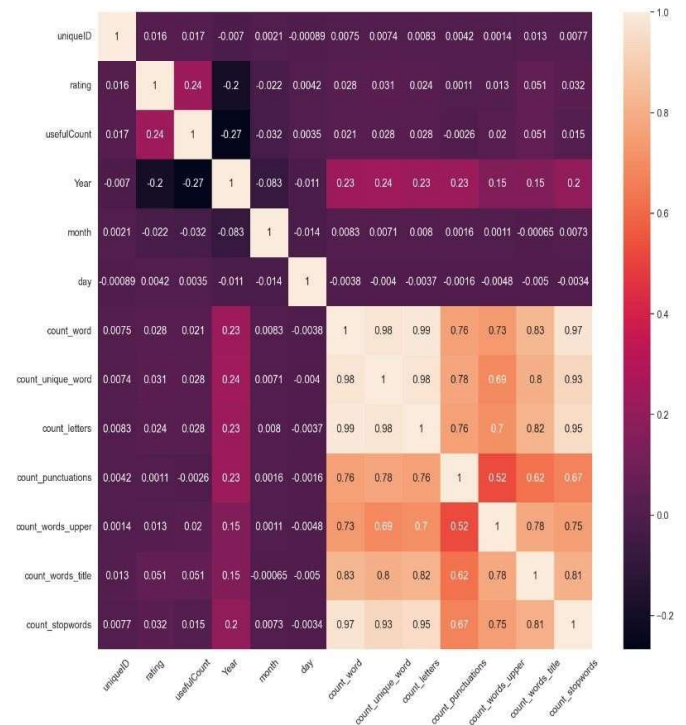


Figure 4: Heatmap for feature selection

Training and testing:

The dataset after feature selection is considered for the process of training and splitting. For this, 80% of the data was used for training purposes while the rest 20% used for evaluation and testing purposes with an addition of shuffling during the training phase. The same setup was used for both ML experiments and Deep Learning experiments as well.

Word Vectorizer:

In this process, 75% of the dataset will transform into numeric format using TF-IDF. The input for this procedure is pre-processed data in the form of tokenized text words, and the output is a vectorized model. The model represents words in a vectorized form which can be used for the working of learning models.

ML Algorithm Setup:

In this process, multiple class classification is the objective given that each review has a probability to be of 3 labels namely positive, neutral, and negative. For the machine learning algorithms, the research is experimented using the following algorithms based on TF-IDF based word vectorization:

1. Logistic Regression.
2. Linear Support Vector Classifier.

For each sentiment labeled by Textblob and Vader separate training using the Machine learning algorithms is performed. Further using the testing data, the model is evaluated.

Ensemble Algorithm Setup:

In this, the combination of both machine learning models namely Logistic Regression and Linear Support Vector Classifier is used. For prediction, the model uses hard voting where each model gets one vote, and the majority class prediction becomes the final prediction. In the case of a tie, it can break it arbitrarily or by using some rule.

DL Algorithm Setup:

In this, the sequential based Deep learning model is used to classify the drugs reviews. LSTM is used on top of TF-IDF. Further, the training is performed based on certain parameters in Deep Learning. The maximum number of words used is 50000, the maximum number of words in each review was parameterized to 250 as the maximum length. Further the epochs used were 5 with a batch size of 64 respectively. This was further used for training and testing the dataset.

Evaluation metrics:

Based on the experimental setup, the evaluation metrics are performed on the TF-IDF based ML and DL algorithms. Further the trained model is evaluated over 20% of the testing dataset. The obtained results are evaluated based on precision, recall and accuracy specifically. Then, moving on, comparisons of combinations of algorithms and models are explained and concluded.

Algorithm	Word Embedding	Lexicon Library
Logistic Regression	TF-IDF	Textblob and Vader
Support Vector Classifier	TF-IDF	Textblob and Vader
Ensemble Model (LR+SVC)	TF-IDF	Textblob and Vader
Deep Learning (LSTM)	TF-IDF	Vader

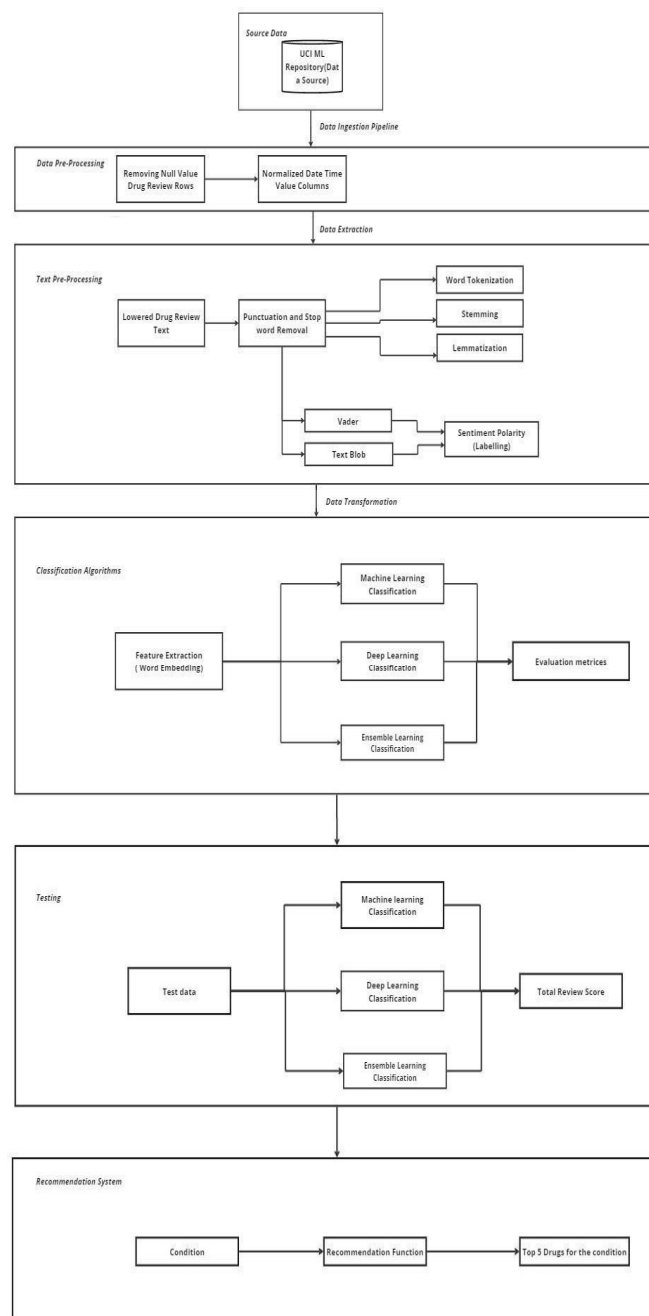


Figure 5: Architecture Diagram

4. RESULTS AND DISCUSSION:

The classification reports of each model are displayed out of which the factor of consideration taken is the accuracy level of each model. In the case of Textblob label, SVC ranks over the benefits of ensemble model by 1 percent whereas in the case of Vader labelling the ensemble model and deep learning model outweighs other classifiers with 0.93 and 0.96 as their accuracy score.

Textblob Label:

Logistic Regression

	precision	recall	f1-score	support
negative	0.95	0.90	0.93	8472
neutral	0.96	0.94	0.95	3348
positive	0.96	0.98	0.97	20259
accuracy			0.95	32079
macro avg	0.95	0.94	0.95	32079
weighted avg	0.95	0.95	0.95	32079

Figure 6: Results on Logistic Regression

Linear Support Vector Classifier

	precision	recall	f1-score	support
negative	0.95	0.93	0.94	8458
neutral	0.97	0.95	0.96	3284
positive	0.97	0.98	0.98	20337
accuracy			0.97	32079
macro avg	0.96	0.96	0.96	32079
weighted avg	0.97	0.97	0.97	32079

Figure 7: Results on SVC

Ensemble Model (LR+LinearSVC)

	precision	recall	f1 score	support
negative	0.94	0.94	0.94	8442
neutral	0.96	0.95	0.95	3322
positive	0.97	0.98	0.97	20315
accuracy			0.96	32079
macro avg	0.96	0.96	0.96	32079
weighted avg	0.96	0.96	0.96	32079

Figure 12: Results on Ensemble Model

Vader Label:

Logistic Regression

	precision	recall	f1 score	support
Negative	0.92	0.95	0.93	14757
Neutral	0.95	0.49	0.65	1971
Positive	0.92	0.95	0.94	15751
accuracy			0.92	32079
macro avg	0.93	0.80	0.84	32079
weighted avg	0.92	0.92	0.92	32079

Figure 8: Results on Logistic Regression

Linear Support Vector Classifier

	precision	recall	f1-score	support
Negative	0.92	0.96	0.94	14601
Neutral	0.93	0.48	0.63	2010
Positive	0.93	0.96	0.94	15468
accuracy			0.93	32079
macro avg	0.93	0.80	0.84	32079
weighted avg	0.93	0.93	0.92	32079

Figure 9: Results on SVC

Ensemble Model (LR+LinearSVC)

	precision	recall	f1 score	support
Negative	0.93	0.97	0.94	14671
Neutral	0.94	0.51	0.66	1949
Positive	0.95	0.94	0.94	15459
accuracy			0.93	32079
macro avg	0.93	0.81	0.85	32079
weighted avg	0.93	0.93	0.92	32079

Figure 10: Results on Ensemble Model

LSTM – Deep Learning

669/669 [=====] - 40s 60ms/step - loss: 0.1114 - accuracy: 0.9642
Test set
Loss: 0.111
Accuracy: 0.964

Figure 11: Results on LSTM

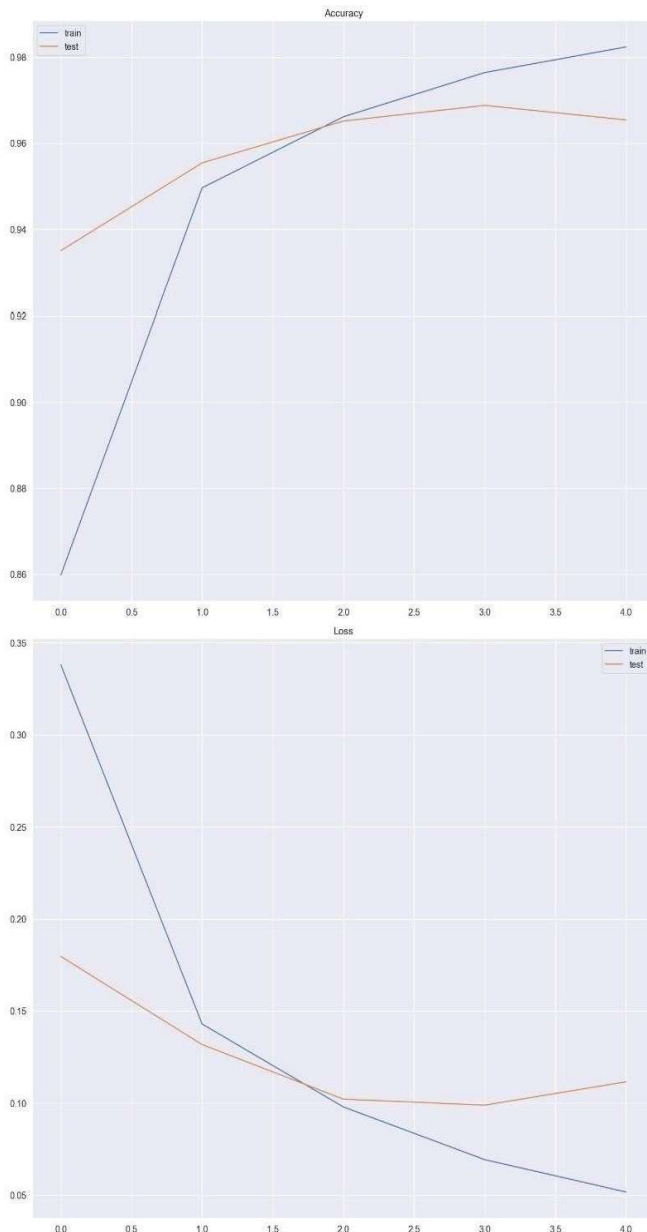


Figure 13: Train and Test evaluation analysis on LSTM Model

5. CONCLUSION AND FUTURE WORK:

The goal of this work is to use aspect-based sentiment analysis with the Textblob and Vader NLP packages to calculate the polarity score in terms of subjectivity and polarity. The UCI repository's drug review dataset is employed. The Textblob approach has higher accuracy ratings than the Vader method in the machine learning scenario. However, with a 96.4% accuracy, LSTM-based modeling performs better than Vader and Text Blob-based categorization. Textblob generates polarity scores that range from 1 to 1, and subjectivity values that range from 0.0 to 1.0.

On the other hand, Vader analysis observes the polarity scores as negative for the remaining values, positive when the score is $\leq (-0.05)$, and neutral when the score is $\geq (0.05)$.

An expert recommendation system could help medical practitioners make therapeutic decisions by providing a comprehensive grasp of the adverse effects and difficulties that patients face. The current system is composed of lexicon-based systems with the addition of various learning methodologies that could have increased performance through greater domain knowledge comprehension. In this case, providing condition-specific information in the evaluations could improve overall efficiency. It increases pharmacovigilance for the given medications and highlights fine-tuned potential drugs by using sentiment analysis on patient input. Other word embeddings, such as word2vec, can be used to gain a better perspective.

Deep learning models require more computer power, which might be lowered. For further implementation, the large language models such as BERT deployed for analysis and classification through which you can even get intricate hidden patterns in the text.

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